

CANDIDATE TAKE-HOME EXERCISE

AI-Powered Live Commerce Platform

Solutions Architect - Design, Build & Explain

The objective

Design and build a working prototype that demonstrates how you translate a customer problem into an end-to-end real-time solution. We are evaluating your architecture, product judgment, customer journey thinking, implementation approach, and ability to explain trade-offs - not your ability to memorize SDK syntax.

What makes this exercise different

- You are not being given a prescribed architecture or a mandatory list of Agora products.
- You should decide which real-time, AI, messaging, recording, transcription, and application components are appropriate for the experience.
- You are expected to distinguish clearly between capabilities powered by Agora and business application logic that the customer's development team would own.
- A functional prototype is expected, but production-grade polish is not.

AI-assisted development is allowed

You are free to use Codex, Claude Code, Cursor, GitHub Copilot, ChatGPT, or other AI-assisted development tools. You may also use frameworks, starter templates, open-source libraries, and third-party services where appropriate. Be prepared to explain the resulting code, architecture, and design decisions.

1. Business Scenario

A traditional multi-category e-commerce company currently operates a web-based shopping platform selling products across categories such as electronics, apparel, cosmetics, home, and lifestyle. The company now wants to evolve the existing website into an interactive live-commerce destination.

Scope of the build

Build a web application only. Android and iOS applications are out of scope. The solution should first behave like a functional e-commerce website and then add live-commerce and voice-AI experiences on top of that foundation.

Path 1 - Live commerce on the website

- Any approved seller, brand representative, or host can start a live session on the platform to demonstrate, promote, or sell products.
- Customers can discover scheduled, currently live, and previously recorded sessions from the website.
- During a live session, viewers can watch the host and interact in real time through chat and other engagement mechanisms you consider appropriate.
- Recorded sessions should remain discoverable so users can replay product demonstrations after the live event has ended.

Path 2 - Voice AI shopping assistance

- Live Sessions: Privately interact with Voice AI without leaving the live shopping experience.
- Recorded Sessions: Use the same Voice AI experience while watching recorded content.
- Website Shopping: Use Voice AI while browsing or skip traditional browsing entirely - describe requirements conversationally and let the AI discover, narrow down, compare, and recommend relevant products.
- The AI assistant has access to the product and commerce information available through the e-commerce application and should use that information to answer customer questions.
- The experience should be voice-first rather than requiring the viewer to type questions to the AI assistant.

Questions the AI should be able to handle

- Product specifications, features, variants, and functionality.
- Comparison with other similar products available on the website - for example price, specifications, capabilities, ratings, or other useful attributes.
- Availability for a customer-entered PIN code or delivery location.
- Available payment options such as credit card, UPI, or cash on delivery.
- Product pricing, offers, and other information exposed by the commerce application.

Commerce actions

- When the customer expresses purchase intent, the AI assistant should be able to help select the required product or variant and add it to the cart.
- The web application should then present the relevant purchase options so the user can continue toward checkout.
- Payment processing itself may be mocked; the important requirement is demonstrating the application workflow and the handoff from AI conversation to commerce action.

Live-session business rule

Products purchased during an active live session receive a 20% live-session discount. The AI assistant should proactively make the viewer aware of this offer when relevant. Once the live session ends, the 20% discount is no longer valid. Your application must model this as business logic rather than hard-coded presentation text only.

Illustrative customer journey

#	Stage	Expected experience
1	Browse	A customer visits the web store, browses multiple product categories, searches products, and views product details.
2	Discover Live	The customer sees upcoming, currently live, and recorded shopping sessions.
3	Join	The customer opens a live session and watches a host demonstrate products.
4	Engage	The customer participates through live chat and other real-time interaction.
5	Ask AI	The customer starts a private voice conversation with the AI shopping assistant.
6	Compare	The assistant answers questions and compares the featured product with alternatives from the website catalog.
7	Check	The assistant can answer availability, PIN-code delivery, pricing, offer, and payment-option questions using application data.
8	Buy	The customer expresses purchase intent; the assistant helps select the product/variant and adds it to the cart.
9	Discount	If the live event is still active, the application applies the 20% live-session offer and clearly communicates it.
10	Checkout	The customer continues through the web application toward checkout using the available purchase options.
11	Replay	After the event, customers can access a recorded session; the live-only discount must no longer apply.

Important

The business journey is defined; the technical architecture is not. You should decide which Agora capabilities, AI components, web technologies, and customer-owned services should power each stage.

2. Your Challenge

Design and build a working web prototype that combines a conventional multi-category e-commerce experience with live shopping and a voice-AI shopping assistant. The prototype should demonstrate both the customer-facing journey and the underlying solution architecture.

Minimum outcomes to demonstrate

- A web storefront containing multiple product categories and product-detail experiences.
- A live-shopping area showing at least scheduled/upcoming, currently live, and recorded sessions.
- A host can start or participate in a live product session and viewers can join it.
- Viewers can interact with the host of the live session through chat.
- A viewer can initiate a real-time voice conversation with an AI shopping assistant during a live session while still tuned into the Live Stream.
- The AI can answer questions from the product catalog and compare products available on the website.
- The AI can use application data to answer questions such as PIN-code availability, pricing, and payment options.
- The AI can trigger an application-side action to add the selected product/variant to the cart.
- The application applies a 20% discount only while the relevant live session is active and removes the eligibility after the session ends.
- Recorded-session playback is available or clearly demonstrated.
- You demonstrate or design how recording and transcription are handled.

Optional extensions

- Live captions or host transcription.
- Reactions, polls, synchronized product cards, or highlighted products during the stream.
- Searchable transcript or session summary for recorded events.
- Multi-language voice assistance with captions.
- Authentication, seller/host roles, wishlists, recommendations, or basic order history.
- Moderation, analytics, observability, or personalized offers.

3. Architecture & Solutioning Expectations

A central part of the exercise is demonstrating where Agora should be used and where conventional e-commerce application logic should remain customer-owned. Your architecture should make these boundaries explicit.

Your architecture should answer

- Which Agora capabilities would you use - and why?
- Would you use Agora Conversational AI Engine, Agora Studio, or another approach for the AI shopping assistant? Explain the choice.
- How would you design the host-to-audience media topology for the prototype and for a large-scale production deployment?
- Which capabilities belong in the customer application layer?
- How will the AI assistant securely access product information and invoke business APIs/tools such as comparison, inventory lookup, and add-to-cart?
- How will the application determine whether a live session is active and enforce the 20% live-session discount correctly in the UI and during interaction with a prospective buyer?
- How will state be shared between the live experience, the AI conversation, and the shopping cart?
- How would recorded sessions, transcription, privacy, consent, and retention be handled?
- What changes when the platform must support very high viewer concurrency and many simultaneous live sellers?
- How would you monitor and troubleshoot failures across Agora, AI providers, and customer-owned web/application services?

Technology stack

Choose the web frontend, backend, database, AI model/provider, deployment approach, and supporting services you believe are appropriate. There is no mandatory stack. We care more about the rationale, integration quality, and system boundaries than the specific framework selected.

Pragmatic implementation is encouraged

You may use mock inventory, product, delivery-serviceability, payment, or checkout services where appropriate. The storefront and commerce flows should be convincing enough to demonstrate the journey, but you do not need to build a production commerce engine. Prioritize architecture, real-time integration, voice AI, and business logic.

5. Expected Submission

Please submit a single repository or shareable project containing the prototype and supporting documentation. The submission should be easy for another engineer to understand and run.

Deliverable	What we expect
Working prototype	A functional implementation of the core journey.
README	Setup instructions, prerequisites, configuration, and how to run the demo.
Architecture diagram	Show the major components, data/media flows, Agora services, AI services, and customer-owned application services.
Customer journey	Show the key user states across storefront browsing, live-session discovery, live interaction, voice-AI assistance, commerce action, discount handling, checkout, and replay.
Responsibility mapping	Clearly identify what is powered by Agora versus what is implemented as customer/business application logic.
Technology rationale	Briefly explain your chosen frontend, backend, data, AI, and deployment stack.
Production considerations	Call out major scalability, reliability, security, privacy, observability, and cost considerations.
Demo walkthrough	Be prepared to present the solution and explain the key implementation decisions.

6. What We Will Evaluate

We are interested in the quality of your decisions and reasoning as much as the final prototype. Your discussion should make it easy for us to understand why you made each major choice.

Solution architecture: Correctness of the overall design, system boundaries, media/data flows, scalability, and production thinking.

Agora product judgment: How effectively you identify and apply relevant Agora capabilities without forcing Agora into areas better handled by the application layer.

Product & customer thinking: Whether the journey is coherent, useful, and designed around an actual customer outcome.

Application engineering: Quality of the frontend/backend integration, APIs, state management, data model, and business logic.

Implementation quality: Whether the core prototype works reliably and the codebase is understandable.

Communication & trade-offs: How clearly you explain alternatives, constraints, compromises, and what you would change for production.

During the discussion

Be prepared for follow-up changes to the problem - for example, higher concurrency, different AI providers, PSTN access, stricter privacy requirements, or a change to the live-session discount and checkout rules. We want to understand how you adapt the architecture when customer requirements change.

Build the experience. Explain the architecture. Defend the trade-offs.

Good luck.